INTRODUCTION TO CARBON CHEMISTRY 1 — 17

Organic chemistry

Organic chemistry is the study of **carbon compounds**. At one time, the term 'organic compound' meant a compound that was found in plant or animal materials, such as sugars, fats and proteins. All of these compounds contain carbon. Now the term organic compounds is used for almost all carbon compounds, whether they come from plants and animals or are made in a laboratory. However, compounds such as carbonates ($-CO_3^{2-}$) and carbon dioxide are not usually described as organic compounds.

1 What is the formula for carbon dioxide?

About 5 million chemical compounds are known to occur on Earth. Over 4 million of these are carbon compounds. However, the Earth contains less than 0.01 % by mass of carbon. This huge discrepancy between the proportion of compounds that contain carbon and the proportion of carbon in the Earth indicates that carbon atoms have some very unusual bonding properties. The bonding in most organic substances is covalent.

2 Does covalent bonding involve sharing electrons or the transfer of electrons?

Carbon is in Group IV of the Periodic Table and has four electrons in the outer shell. Because of this, it tends to form *four* covalent bonds. The four electrons available in the outer shell also allows for a much larger number of possible combinations with other atoms. This enables carbon to form many different molecules and explains why it is present in so many compounds.

The bonds form a three-dimensional tetrahedral structure that is often simplified to a two-dimensional view as shown in figure 1, which illustrates the structural formula of a methane molecule. The **structural formula** shows how the atoms are arranged in a molecule (think of a tripod). The **molecular formula** shows the number and type of elements present in each molecule.



Figure 1

3 What is the molecular formula of methane?

The first diagram in figure 1 shows two H atoms in the plane of the sheet of paper (solid lines), one hydrogen atom behind the plane of the paper (dotted lines) and one hydrogen atom coming forward out of the plane of the paper (solid triangle). Methane is a hydrocarbon.

4 Hydrocarbons consist of only two elements. What are these elements?

Alkanes Methane is the first in a series of structurally similar hydrocarbons called **alkanes**. All alkanes have a name ending in '-ane'. The next alkanes after methane are ethane, propane and butane. Their structures are shown in figure 2. Each straight line in figure 2 stands for a single, covalent bond (i.e. a shared pair of electrons). Remember that these are *not* flat, two-dimensional molecules, but three-dimensional (zigzag) molecules.

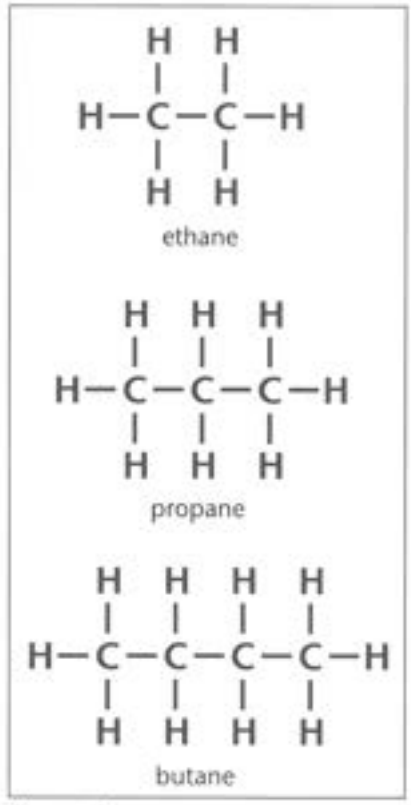


Figure 2

5 Give the formula for each alkane.

a ethane **b** propane **c** butane

Saturated means 'to hold as much as can be held'. Alkanes are 'saturated' hydrocarbons; that is, all carbon-to-carbon bonds are single bonds and there are no available electrons for 'extra' atoms to bond with. Carbon has a unique ability for **concatenation**; that is, it can form long chains (concatenation means *to link*). The alkanes are a **homologous series**. This means they have similar structures. They differ by a single repeated unit, CH_2, and show a gradual change in physical properties according to chain length. At room temperature, the first four alkanes are gases, the next twelve are liquids and the remaining alkanes are solids.

6 The next members of the alkanes are pentane, hexane, heptane, octane, nonane and decane. Give the structural and molecular formulae for each of these molecules.

7 Explain what is meant by the statements: 'Alkanes have the general formula of C_nH_{2n+2}.'

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Large deposits of alkanes are quite widespread as fossil fuels (e.g. coal, oil and natural gas). Longer chain alkanes are mainly isolated by the fractional distillation of crude oil (petroleum). Alkanes are relatively unreactive. One major reaction alkanes undergo is combustion.

8 Do alkanes conduct electricity? Explain your answer. (Hint: think about bonding.)

9 What is another name for *combustion*?

10 Given that the products of the combustion of any hydrocarbon (in excess oxygen) are carbon dioxide and water*, write a balanced formula equation for the combustion of butane.
(* Note that if insufficient oxygen is present, hydrocarbons burn to form water and carbon and/or carbon monoxide.)



Figure 3

Substituted alkanes The hydrogen in alkanes can be replaced by other elements, especially halogens. Figure 3 shows the structural formula of monochloromethane (CH_3Cl).

11 Give the molecular and structural formula of each compound.

a dichloromethane **b** trichloromethane **c** tetrachloromethane

Alkenes The alkenes are another family of hydrocarbons. They have the general formula C_nH_{2n}. The main differences between alkenes and alkanes are that alkenes have at least one carbon–carbon double bond (two shared pairs of electrons) and they are much more reactive than alkanes.

Alkenes are called 'unsaturated' hydrocarbons. To name an alkene, the ending '-ene' is added to the root derived from the corresponding saturated hydrocarbon.

12 How does the general formula for alkenes compare with that of alkanes?

13 Suggest why alkenes are more reactive than alkanes.

Figure 4 shows the structural formulae of ethene, propene and butene (sometimes called ethylene, propylene and butylene).

14 Explain why methene cannot exist.

15 Give the molecular formula for each alkene.

a ethene **b** propene **c** butene

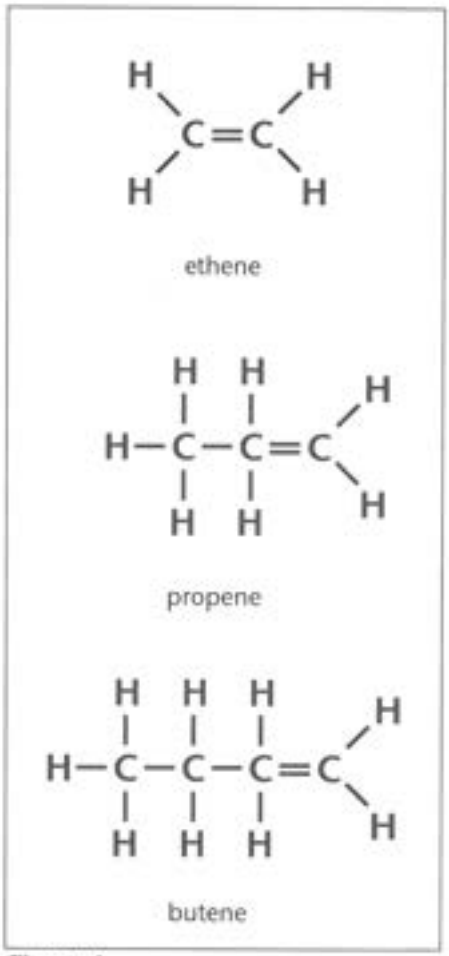


Figure 4

Alkynes Carbon to carbon *triple* bonds can also exist and these bonds are found in alkynes. Alkynes are very reactive hydrocarbons. To name an alkyne, add the ending '-yne' to the root derived from the corresponding saturated hydrocarbon. Figure 5 shows the structural formulae of ethyne and propyne. Like alkanes and alkenes, alkynes also form a homologous series.

Ethyne is the most common alkyne. Its common name is *acetylene*. With oxygen, it burns at a very high temperature and is used for welding metals.

16 Suggest why alkynes are more reactive than alkenes.

17 Give the molecular formula for each alkyne.

a ethyne **b** propyne

18 Give the name, structural formula and molecular formula for the next alkyne in the series after ethyne and propyne.

19 Write a balanced equation for the combustion of ethyne.

20 Give the general formulae for alkanes, alkenes and alkynes.

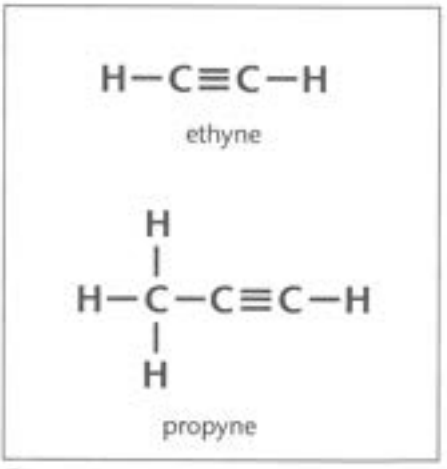


Figure 5

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